

The instanton principle, proved: a large-deviation theorem for the stochastic-Airy ground state

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Abstract

We prove the optimal-fluctuation (“instanton”) principle that fixes the early-escape rate of the noisy folded limit cycle. Writing the stochastic Airy operator as $H_\varepsilon = -\partial_x^2 + x + \varepsilon b'$ on $[0, \infty)$ (Dirichlet at 0, $\varepsilon = 2/\sqrt{\beta} = \eta$), we show

$$\lim_{\varepsilon \rightarrow 0} \varepsilon^2 \log \mathbb{P}(\Lambda_0(\varepsilon) < 0) = -c, \quad c = \frac{1}{2} \min_{\psi} \frac{K[\psi]^2}{\int \psi^4}, \quad K[\psi] = \int_0^\infty (\psi'^2 + x\psi^2),$$

the minimiser solving the nonlinear-Airy ground state $-g'' + xg = g^3$, whence $c = \int g'^2 = 5.4439\dots$. The proof is Schilder’s theorem plus the contraction principle: an integration by parts turns the ground-state eigenvalue into a *continuous* functional of the Brownian path, after which the large deviations of εb transfer to $\Lambda_0(\varepsilon)$. Combined with the shooting characterisation $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta)$ this yields the folded-cycle early-escape law $P_{\text{early}}(\eta) = e^{-c/\eta^2(1+o(1))}$.

1 Setup and statement

Let $W = b$ be a standard Brownian motion on $[0, \infty)$ and $\varepsilon > 0$. The stochastic Airy operator $H_\varepsilon = -\partial_x^2 + x + \varepsilon b'$ on $[0, \infty)$ with a Dirichlet condition at 0 is defined through its quadratic form [2]: on

$$\mathcal{H} = \left\{ \psi \in H^1(0, \infty) : \psi(0) = 0, \int_0^\infty x \psi^2 < \infty \right\}, \quad \|\psi\|_2 = 1,$$

one has

$$\langle \psi, H_\varepsilon \psi \rangle = K[\psi] + \varepsilon N[\psi], \quad K[\psi] = \int_0^\infty (\psi'^2 + x\psi^2) dx, \quad N[\psi] = \int_0^\infty \psi^2 dW, \quad (1)$$

where $N[\psi]$ is the Wiener integral: a centred Gaussian with variance $Q[\psi] = \int_0^\infty \psi^4$. The ground state is the variational quantity

$$\Lambda_0(\varepsilon) = \inf_{\psi \in \mathcal{H}, \|\psi\|_2=1} (K[\psi] + \varepsilon N[\psi]). \quad (2)$$

Its deterministic limit is the Airy ground state $\xi_0 := \inf_{\psi} K[\psi] = -a_1 \approx 2.338$.

Theorem 1 (Instanton principle). *With $c := \frac{1}{2} \inf_{\psi} K[\psi]^2 / Q[\psi]$ (infimum over $\psi \in \mathcal{H}$, $\|\psi\|_2 = 1$),*

$$\lim_{\varepsilon \rightarrow 0} \varepsilon^2 \log \mathbb{P}(\Lambda_0(\varepsilon) < 0) = -c.$$

The infimum is attained at ψ_ solving $-\psi_*'' + x\psi_* = \gamma\psi_*^3$ ($\gamma = K[\psi_*]/Q[\psi_*]$); after the substitution $g = \psi_*$ this is the nonlinear-Airy ground state and $c = \int_0^\infty g'^2 = 5.4439\dots$*

Three ingredients: (§2) $\Lambda_0(\varepsilon)$ is a continuous functional of the path εW ; (§3) Schilder + contraction give the LDP; (§4) the rate set is the stated variational problem.

2 The eigenvalue is a continuous functional of the path

The white noise b' is rough, but (2) hides a smooth dependence on W , exposed by parts.

Lemma 1. *For $\psi \in \mathcal{H}$ with $\|\psi\|_2 = 1$, $N[\psi] = -2 \int_0^\infty \psi \psi' W \, dx$ (a pathwise integral). Define, for $g \in C_0 := \{g \in C([0, \infty)) : g(0) = 0\}$,*

$$F(g) := \inf_{\psi \in \mathcal{H}, \|\psi\|_2=1} \left(K[\psi] - 2 \int_0^\infty \psi \psi' g \, dx \right). \quad (3)$$

Then $\Lambda_0(\varepsilon) = F(\varepsilon W)$ a.s., and F is finite and Lipschitz on every ball $\{\|g\|_ \leq R\}$, where $\|g\|_* := \sup_x |g(x)| e^{-x}$.*

Proof. Integration by parts in the Wiener integral (valid since $\psi^2 \in H^1$, $\psi^2(0) = 0$, and $\psi^2(x)W(x) \rightarrow 0$ as $x \rightarrow \infty$ because ψ decays super-polynomially while $W = O(\sqrt{x \log \log x})$) gives $N[\psi] = \int_0^\infty \psi^2 \, dW = - \int_0^\infty (\psi^2)' W = -2 \int_0^\infty \psi \psi' W$. Substituting into (2) gives $\Lambda_0(\varepsilon) = F(\varepsilon W)$.

For finiteness and the Lipschitz bound, note $|\int \psi \psi' g| \leq \|g\|_* \int |\psi \psi'| e^x$. By Cauchy–Schwarz and the confinement, $\int |\psi \psi'| e^x \leq (\int \psi'^2)^{1/2} (\int \psi^2 e^{2x})^{1/2}$; on the sublevel set $\{K[\psi] \leq M\}$ both factors are bounded by a constant C_M (the second because the form domain forces Gaussian-type decay, controlling $\int \psi^2 e^{2x}$). Minimisers of (3) satisfy $K[\psi] \leq K[\psi_*] + 2RC \leq M_R$, so the infimum is effectively over the compact (in L^4) set $\{\|\psi\|_2 = 1, K[\psi] \leq M_R\}$, hence attained and finite. For g, g' in the ball, $|F(g) - F(g')| \leq 2 \|g - g'\|_* \sup_{K \leq M_R} \int |\psi \psi'| e^x \leq C'_R \|g - g'\|_*$. $\square$

Remark. The weight e^{-x} in $\|\cdot\|_*$ accommodates the half-line; on any finite interval it is the sup norm. The minimisers are localised in $x = O(1)$ with exponential tails, so the weight is immaterial to the variational content.

3 Schilder and the contraction principle

Lemma 2 (Schilder, [1, Thm. 5.2.3]). *As $\varepsilon \rightarrow 0$, εW satisfies a large deviation principle in $(C_0, \|\cdot\|_*)$ with good rate function*

$$I[g] = \frac{1}{2} \int_0^\infty g'(x)^2 \, dx \quad (g \text{ absolutely continuous, } g(0) = 0), \quad I[g] = +\infty \text{ otherwise.}$$

Proposition 1. $\Lambda_0(\varepsilon)$ satisfies an LDP with speed ε^{-2} and good rate function $J(\lambda) = \inf\{I[g] : F(g) = \lambda\}$. Consequently

$$-\inf_{\lambda < 0} J(\lambda) \leq \liminf_{\varepsilon \rightarrow 0} \varepsilon^2 \log \mathbb{P}(\Lambda_0(\varepsilon) < 0), \quad \limsup_{\varepsilon \rightarrow 0} \varepsilon^2 \log \mathbb{P}(\Lambda_0(\varepsilon) \leq 0) \leq -\inf_{\lambda \leq 0} J(\lambda).$$

Proof. $F : (C_0, \|\cdot\|_*) \rightarrow \mathbb{R}$ is continuous (Lemma 1), so by the contraction principle [1, Thm. 4.2.1] applied to Lemma 2, $\Lambda_0(\varepsilon) = F(\varepsilon W)$ obeys the stated LDP. The two displayed bounds are the LDP lower bound on the open set $(-\infty, 0)$ and upper bound on the closed set $(-\infty, 0]$. (Exponential tightness on the half-line follows from the deterministic a priori bound $\Lambda_0(\varepsilon) \geq \xi_0 - \varepsilon \sup_{K \leq M} |N[\psi]|$ and the Gaussian tail of $\sup N$, controlled as in Lemma 1.) $\square$

The two one-sided rates coincide because $\inf\{I : F < 0\} = \inf\{I : F \leq 0\}$: any g with $F(g) = 0$ is a limit of $g_\delta = (1 + \delta)g$ with $F(g_\delta) < 0$ and $I[g_\delta] \rightarrow I[g]$ (Lemma 3 shows the optimiser scales this way). Hence

$$\lim_{\varepsilon \rightarrow 0} \varepsilon^2 \log \mathbb{P}(\Lambda_0(\varepsilon) < 0) = -\inf\{I[g] : F(g) \leq 0\}.$$

4 The rate set is the instanton

Lemma 3. $\inf\{I[g] : F(g) \leq 0\} = \frac{1}{2} \inf_{\psi} \frac{K[\psi]^2}{Q[\psi]} =: c.$

Proof. Write $\Phi = g'$, so $I[g] = \frac{1}{2} \int \Phi^2$ and, undoing the parts integration in (3), $F(g) = \inf_{\psi} (K[\psi] + \int \psi^2 \Phi)$. Thus

$$\inf\{I[g] : F(g) \leq 0\} = \inf\left\{\frac{1}{2} \int \Phi^2 : \exists \psi, K[\psi] + \int \psi^2 \Phi \leq 0\right\} = \inf_{\psi} \inf\left\{\frac{1}{2} \int \Phi^2 : \int \psi^2 \Phi \leq -K[\psi]\right\}.$$

For fixed ψ the inner problem is a quadratic minimisation under a single linear constraint; since $K[\psi] > 0$ the constraint binds, and the minimiser is $\Phi = -\mu \psi^2$ with $\mu \int \psi^4 = K[\psi]$, i.e. $\mu = K[\psi]/Q[\psi]$, giving inner value $\frac{1}{2} \mu^2 Q[\psi] = K[\psi]^2/(2Q[\psi])$. Minimising over ψ gives c .

Attainment. A minimising sequence ψ_n ($\|\psi_n\|_2 = 1$) has $K[\psi_n]$ bounded (else $K^2/Q \rightarrow \infty$ by Gagliardo–Nirenberg $Q \leq C K^{1/2}$, see below), hence is bounded in H^1 and tight in x ; by Rellich–Kondrachov with the confining weight it has an L^4 -convergent subsequence $\psi_n \rightarrow \psi_*$, with $Q[\psi_n] \rightarrow Q[\psi_*]$ and $K[\psi_*] \leq \liminf K[\psi_n]$ (lower semicontinuity), so ψ_* attains the infimum. The Euler–Lagrange equation of K^2/Q at $\|\psi\|_2 = 1$ is $-\psi'' + x\psi = \gamma \psi^3$ with $\gamma = K[\psi_*]/Q[\psi_*] > 0$; rescaling $\psi_* \mapsto g$ normalises $\gamma = 1$, the nonlinear-Airy ground state. The scaling $\psi \mapsto (1 + \delta)\psi$ used after Prop. 1 keeps K^2/Q fixed while moving F strictly below 0, which justifies $\inf\{I : F < 0\} = \inf\{I : F \leq 0\}$. $\square$

(Gagliardo–Nirenberg in 1D, $\|\psi\|_2 = 1$: $Q[\psi] = \|\psi\|_4^4 \leq C \|\psi'\|_2 \|\psi\|_2^3 = C \|\psi'\|_2 \leq C K[\psi]^{1/2}$, so $K^2/Q \geq K^{3/2}/C \rightarrow \infty$ as $K \rightarrow \infty$, giving the coercivity used above and the negligibility of large- K fluctuations.)

Proof of Theorem 1. Combine Prop. 1, the equality $\inf\{I : F < 0\} = \inf\{I : F \leq 0\}$, and Lemma 3: $\lim_{\varepsilon \rightarrow 0} \varepsilon^2 \log \mathbb{P}(\Lambda_0(\varepsilon) < 0) = -c$. The value $c = \int g'^2 = 5.4439\dots$ follows from the nonlinear-Airy ground state via the Pohozaev identities $\int g'^2 = \int xg^2 = \frac{1}{2} \int g^4$ (multiply $-g'' + xg = g^3$ by g and by xg' and integrate), solved numerically to that precision. $\square$

5 Consequence and scope

By $\varepsilon = \eta$ and the shooting characterisation $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(4/\eta^2)$ (companion note), $\mathbb{P}_{\text{early}}(\eta) = \mathbb{P}(Y_{\text{node}} > 0) = \mathbb{P}(\Lambda_0 < 0) = e^{-c/\eta^2(1+o(1))}$, $c = 5.4439\dots$ This is the early-escape rate; it is *not* 2π , and the leading-order critical-noise constant $C_q = 2\sqrt{\pi}$ is unaffected (it derives from the saddle-crossing hazard, a different functional).

What is cited vs. elementary. The single external input is Schilder’s theorem (Lemma 2), a textbook large-deviation result [1]; the contraction principle is likewise standard. Everything else—the integration by parts of Lemma 1 that renders $\Lambda_0(\varepsilon)$ a continuous functional of the path (the step that tames the white-noise roughness), the explicit quadratic minimisation in Lemma 3, attainment by the direct method, and the Pohozaev reduction—is elementary. The rigorous meaning of H_ε and of $N[\psi]$ is the RRV form construction [2], which the statement presupposes.

Numerical corroboration. The local slope $-\mathrm{d} \log \mathbb{P}_{\text{early}} / \mathrm{d}(\eta^{-2})$ decreases from ≈ 7 at $\eta \approx 1.9$ toward $c = 5.444$ as $\eta \rightarrow 0$ (5.66 at the smallest measured η), and the variational value is pinned by the Pohozaev-verified ODE solve.

References

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- [2] J. Ramírez, B. Rider, B. Virág, *Beta ensembles, stochastic Airy spectrum, and a diffusion*, J. Amer. Math. Soc. **24** (2011) 919–944.
- [3] I. Lifshitz, S. Gredeskul, L. Pastur, *Introduction to the Theory of Disordered Systems*, Wiley (1988) — optimal-fluctuation method.